UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

CHEMISTRY **5070/04**

Paper 4 Alternative to Practical **October/November 2008**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

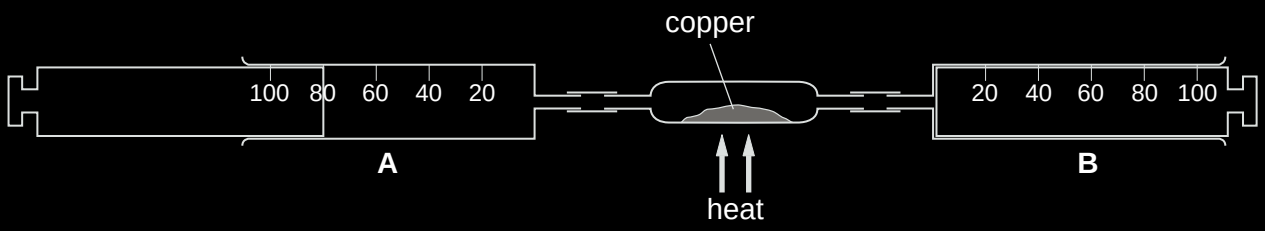
Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.
The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use

This document consists of **17** printed pages and **3** blank pages.

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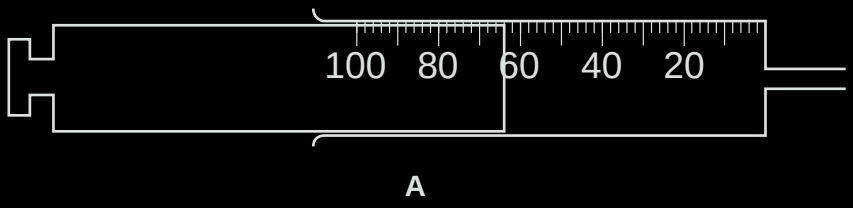
UNIVERSITY of CAMBRIDGE International Examinations



Syringe **A** contained 80 cm^3 of air. The air was forced over heated copper into syringe **B**. The air was then forced back into syringe **A**.

The process was repeated several times until the volume of gas forced back into syringe **A** was constant.

The diagram below shows the volume of gas in syringe **A** after the experiment was finished.



(a) **(i)** Name the major component of the gas remaining in syringe **A**.

.....[1]

(ii) What is the volume of gas remaining in syringe **A**?

.....[1]

(iii) Calculate the percentage of oxygen in the original sample of air.

.....[1]

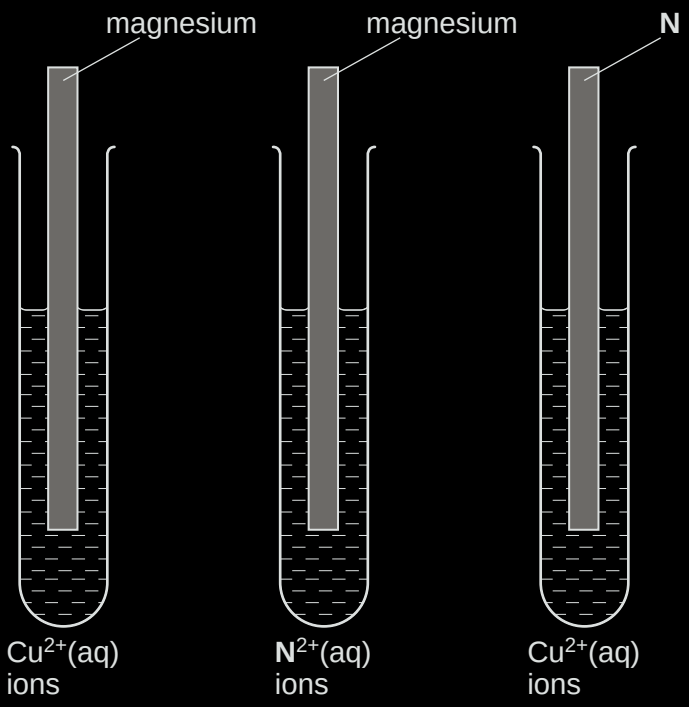
(b) The copper reacted with oxygen in the air to produce copper(II) oxide.

(i) Write the equation for this reaction.

.....[1]

(ii) What colour is copper(II) oxide?

.....[1]



Metal **N** could be

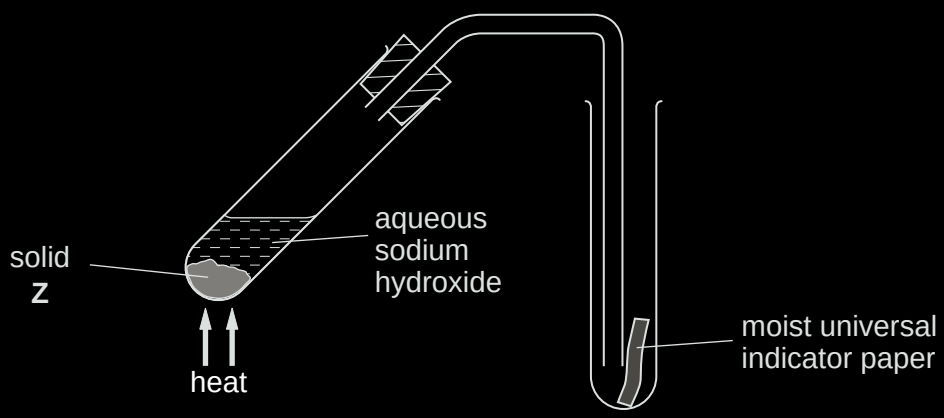
(a) aluminium. ☐

(b) calcium. ☐

(c) silver. ☐

(d) zinc. ☐

[Total: 1]



When the tube was heated, the indicator paper turned blue.
Suggest the identity of **Z**.

(a) aluminium oxide ☐

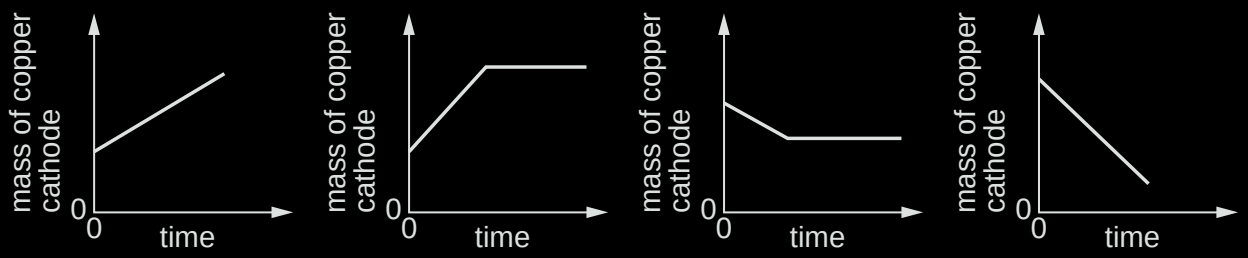
(b) ammonium sulphate ☐

(c) calcium hydroxide ☐

(d) sodium chloride ☐

[Total: 1]

6 A student electrolysed aqueous copper(II) sulphate using copper electrodes.
The electrolysis continued until no further change took place.
Which graph correctly shows how the mass of the copper cathode varied with time.



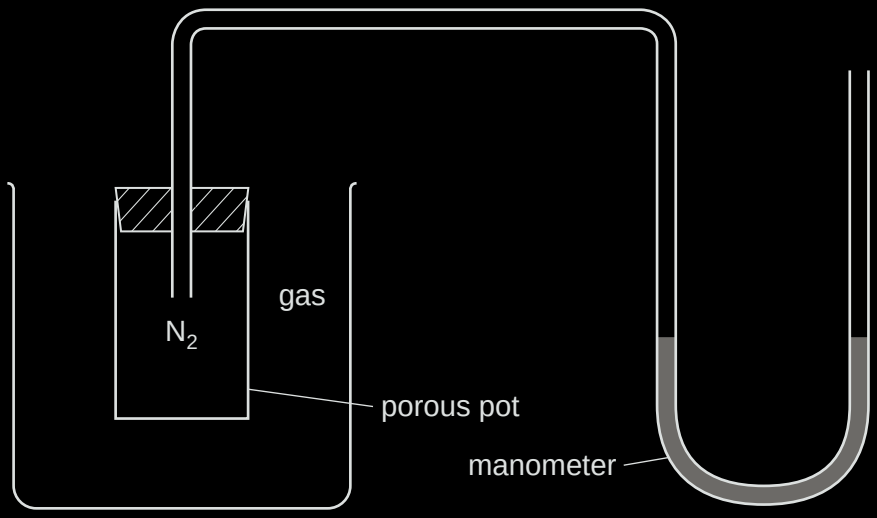
(a) **A** ☐

(b) **B** ☐

(c) **C** ☐

(d) **D** ☐

[Total: 1]



Which gas did **not** cause a change in the level of the manometer?
[A_r: C*l*, 35.5; C, 12; H, 1; O, 16; N, 14]

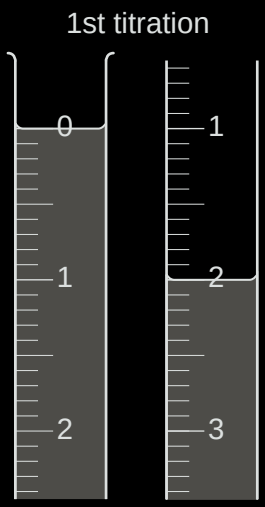
(a) Cl_2 ☐

(b) CO ☐

(c) CH_4 ☐

(d) H_2 ☐

[Total: 1]



(d) After looking at these results the student decided that the hydrochloric acid should be diluted. Explain why he came to this conclusion.

.....[1]

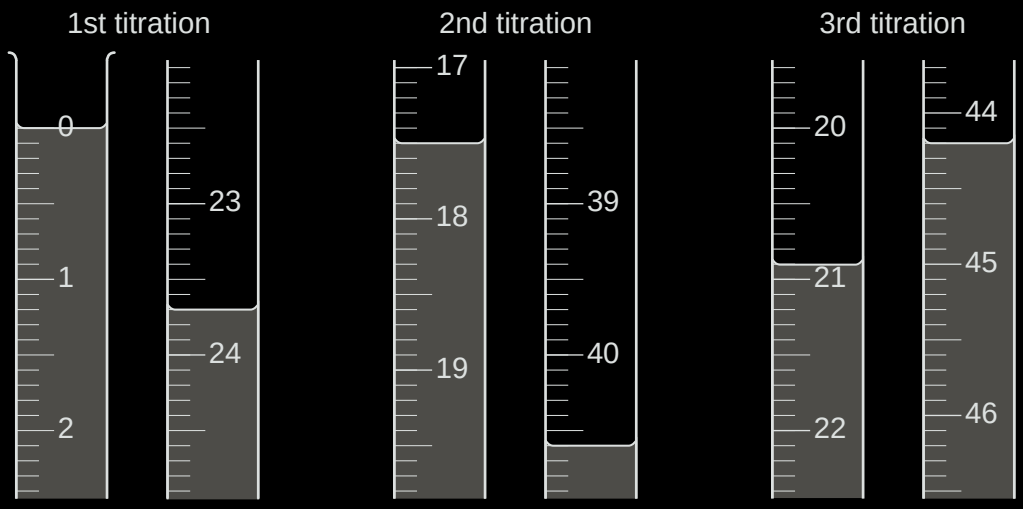
The student transferred 25.0 cm^3 of the acid to a volumetric flask and made it up to 250 cm^3 with distilled water. This was solution **H**.

The student poured away the acid that was left in the burette.

(e) Before refilling the burette with **H**, the burette was washed with two different liquids. Which two liquids did the student use to wash out the burette?

first liquid .....

second liquid.....[2]



(f) Use the diagrams to complete the following table.

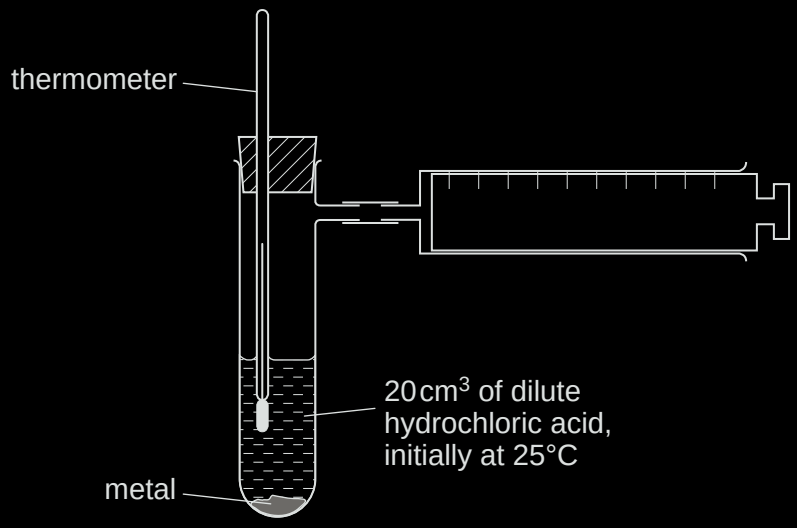
titration number	1	2	3
final burette reading / cm^3			
initial burette reading / cm^3			
volume of **H** used / cm^3			
best titration results (✓)			

Summary
Tick (✓) the best titration results.
Using these results the average volume of **H** used was ..... cm^3.

[4]

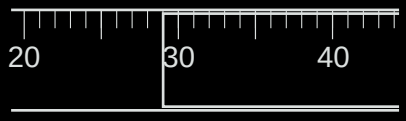
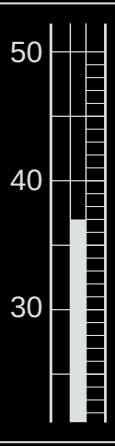
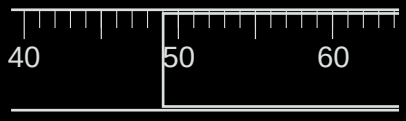
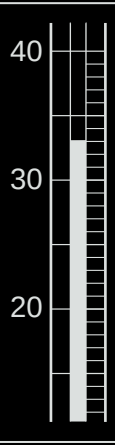
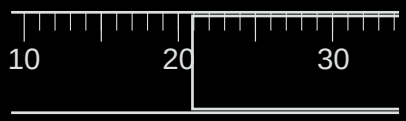
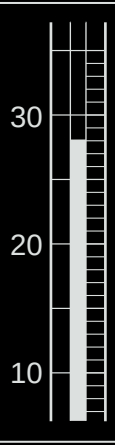
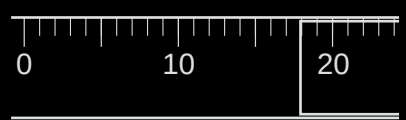
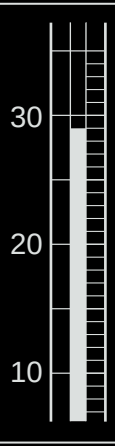
(g) Using your answer to **(b)(ii)** calculate the number of moles of sodium carbonate in 25.0 cm^3 of **G**.

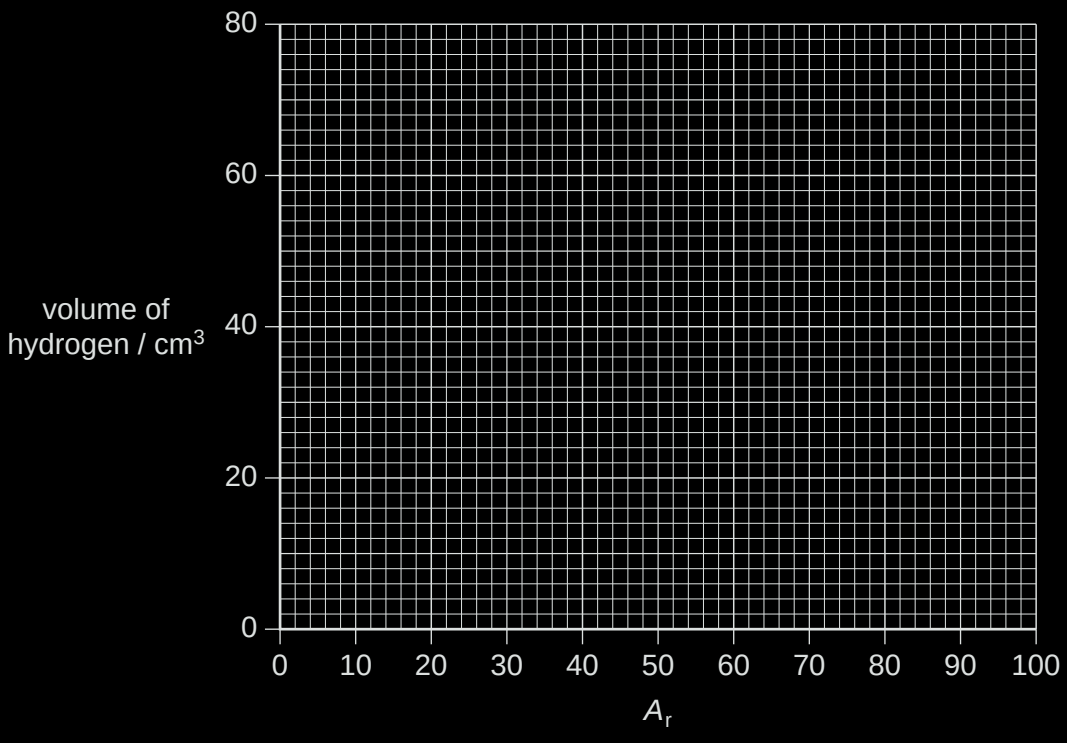
.....moles [1]



Four metals were used, calcium, magnesium, iron and zinc. In each experiment, the initial temperature of the acid was 25°C.
0.05g of the metal was added. When all the metal had dissolved the following measurements were made.

- the volume of hydrogen collected in the syringe
- the highest temperature reached

metal	syringe / cm^3	highest temperature / °C
calcium		
magnesium		
iron		
zinc		



[1]

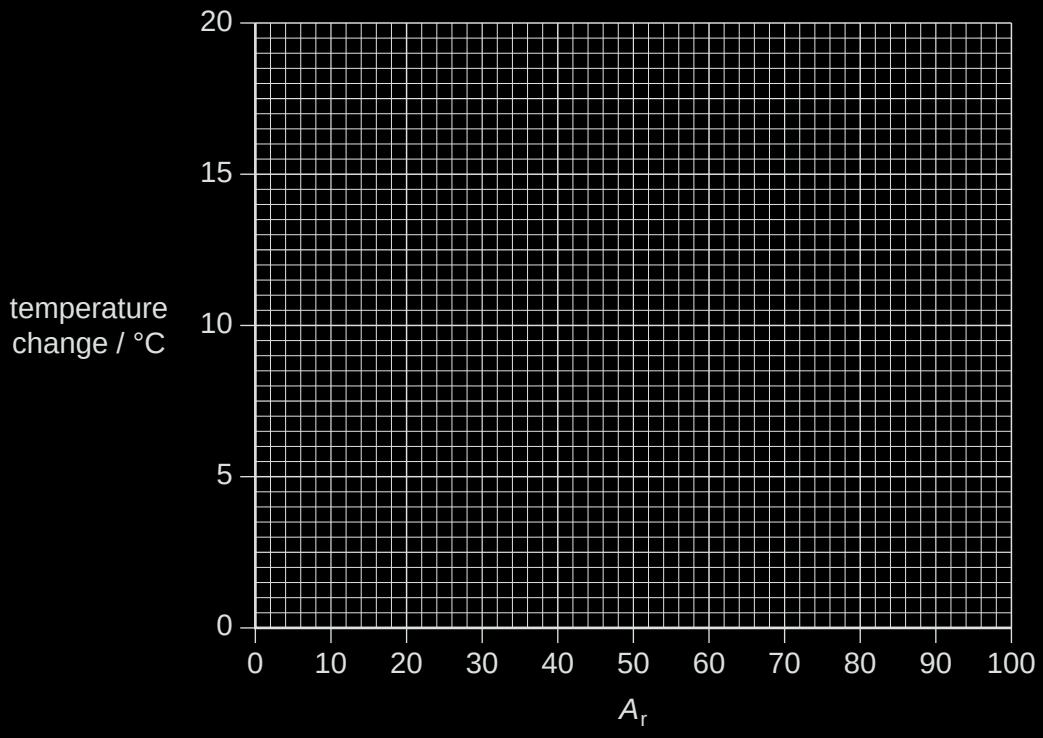
(c) **(i)** By extending the curve, predict the volume of hydrogen that would be produced when 0.05 g of strontium was added to 20 cm^3 of 0.5 mol/dm^3 hydrochloric acid.
[A_r: Sr, 88]

.....cm^3 [1]

(ii) Suggest why the volume of hydrogen produced decreased as the relative atomic mass of the element increased.

.....

..... [1]



[2]

(e) Explain why the temperature rise for the addition of strontium cannot be predicted from the graph.

.....

..... [1]

(f) Suggest why copper was not used in this experiment.

.....

..... [1]

[Total: 10]